

1) Explain your project

Sample Answer:

I worked on a cloud-based infrastructure project where we deployed a scalable web application on **AWS** using Terraform and Kubernetes.

- Used Terraform to provision:
 - VPC, subnets, EC2, security groups
- Deployed Kubernetes cluster (EKS)
- Application deployed using Kubernetes manifests
- CI/CD pipeline automated using Jenkins/GitHub Actions
- Monitoring via CloudWatch & Prometheus

Goal:

- Infrastructure as Code
 - Automated deployments
 - High availability & scalability
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2) Explain the whole CI/CD pipeline used in your project

Pipeline Flow:

1. Developer pushes code to GitHub
 2. CI tool (Jenkins/GitHub Actions) triggers pipeline
 3. Build stage:
 - Compile code
 - Build Docker image
 4. Test stage:
 - Unit tests
 5. Push image to Docker Hub / ECR
 6. CD stage:
 - Deploy to Kubernetes using `kubectl` or Helm
 7. Monitoring & alerts configured
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3) What is Terraform, and how can you manage Terraform state?

Terraform:

- Infrastructure as Code tool used to provision cloud resources

State Management:

- Use remote backend (S3)
- Enable locking (DynamoDB)
- Enable versioning
- Encrypt state file
- Avoid local `.tfstate`

4) Difference between `terraform.tf` and `providers.tf`

File	Purpose
<code>terraform.tf</code>	Backend configuration (state storage)
<code>providers.tf</code>	Provider details (AWS, Azure, etc.)

Example:

- `terraform.tf` → S3 backend
- `providers.tf` → AWS region & credentials

5) What are variables in Terraform?

Variables are used to make code dynamic and reusable.

Types:

- Input variables
- Output variables
- Local variables

Example:

```
variable "instance_type" {
  default = "t2.micro"
}
```

6) What is CloudTrail, and how can you set it up?

AWS CloudTrail:

- Logs all API activity in AWS account

Setup Steps:

1. Go to AWS Console
2. Open CloudTrail
3. Create trail
4. Select S3 bucket for logs
5. Enable logging

Use Cases:

- Security auditing
 - Compliance
 - Troubleshooting
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7) What issues do you face in your project?

Common issues:

- Terraform state conflicts
 - CI/CD pipeline failures
 - Kubernetes pod failures
 - Permission issues (IAM roles)
 - Network misconfiguration
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8) What are Placement Groups in EC2?

Placement groups control how EC2 instances are placed.

Types:

- **Cluster** → Low latency
 - **Spread** → High availability
 - **Partition** → Big data workloads
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9) How to create EC2 using Terraform?

```
resource "aws_instance" "example" {  
  ami           = "ami-123456"  
  instance_type = "t2.micro"  
}
```

Steps:

1. Write config
2. terraform init
3. terraform plan

4. terraform apply

10) If duplicate resources are created, how will you resolve it?

Solutions:

- Check state file (terraform state list)
- Import existing resource:

```
terraform import aws_instance.example <id>
```

- Remove duplicates from code/state
 - Use unique naming
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11) Explain CrashLoopBackOff error

- Pod keeps restarting continuously

Causes:

- App crash
- Wrong config
- Missing dependency

Fix:

```
kubectl logs <pod>
kubectl describe pod <pod>
```

12) OOMKilled error

- Container killed due to **Out Of Memory**

Causes:

- Memory limit exceeded

Fix:

- Increase memory limits
 - Optimize application
-

13) How do you provide volume to any node?

In Kubernetes:

- Use **Persistent Volume (PV)** and **Persistent Volume Claim (PVC)**

```
volumeMounts:
- mountPath: /data
  name: my-volume
```

14) How kubectl interacts with Kubernetes

- kubectl talks to **API Server**
 - Uses kubeconfig file
 - API server communicates with:
 - etcd
 - scheduler
 - kubelet
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15) If I want to add a new node to cluster, how will I do that?

For kubeadm:

```
kubeadm join <master-ip> --token <token>
```

For EKS:

- Increase node group size
- Use Auto Scaling Group

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Q
Count and for_each in Terraform

Q
3 resources in state and 2 resources added manually in AWS

Q
Two developers apply Terraform at the same time

Q

How to avoid terminating EC2 in Terraform

Q

Security Group updated manually in AWS but not present in state file

Q

Copy and add diff in Terraform

Q

Can we create image from container (reverse process)

Q

Jenkins secret manager

Q

Explain Jenkins CI/CD pipeline

Q

How to check memory usage using shell scripting

Q

How to kill top 5 processes using script

Q

Install Nginx using Terraform